Project – Milestone 1

Contents

Group members 1

Member Roles 1

Communication Method 2

Gantt Chart 3

GitHub Link 4

Solutions 4

Problem Statement 1 4

Problem Statement 2 5

Problem Statement 3 7

Group members

Austin Hall
email: austinhall@ufl.edu
GitHub: aahall05

Stephen Kramer
email: kramers@ufl.edu
GitHub: sokramerz

Dajana Seitllari
Email: seitllarid@ufl.edu
Github: seitllarid

Richard Huang
Email: rhuang3@ufl.edu
Github: richardhuang12

Member Roles

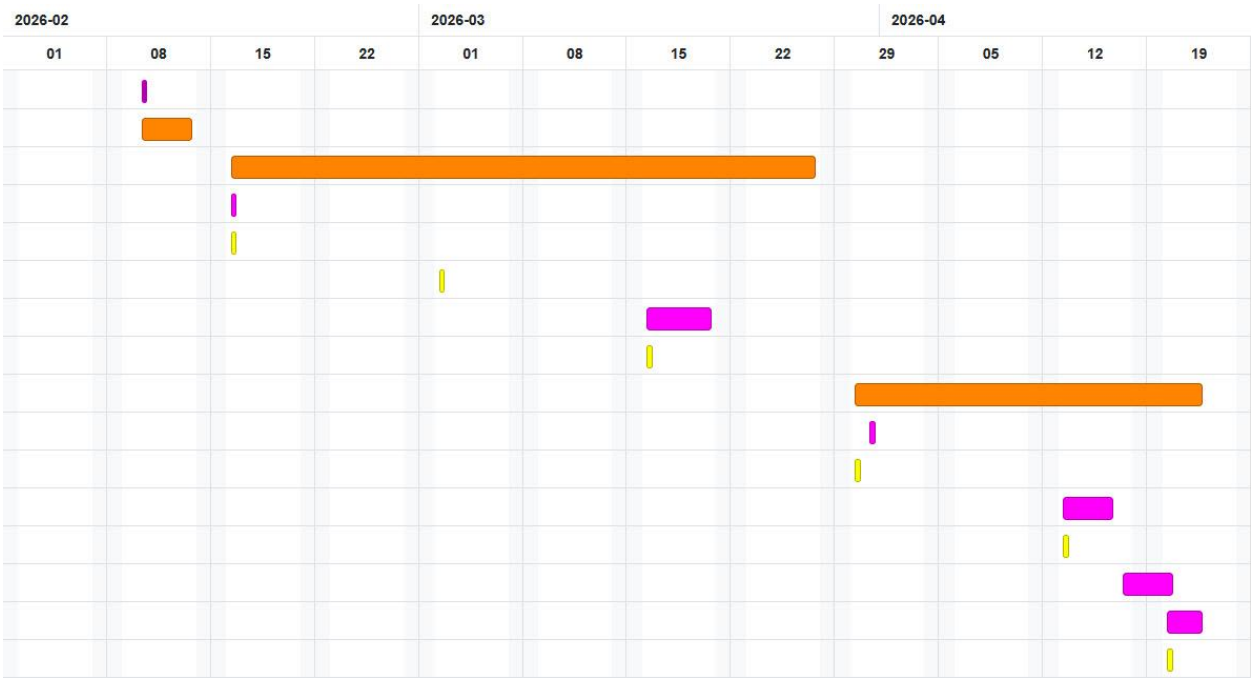
Austin Hall – Product Manager

Stephen Kramer – Scrum Master

Communication Method

The group created a Discord server to communicate. The discord contains our emails and phone numbers for emergency situations.

Project Gantt Chart



ID	Task Name	Start	End	Duration	Color
1	Project file and Github creation	2026-02-10	2026-02-10	1 day	
2	Milestone 1	2026-02-10	2026-02-13	4 days	
3	Milestone 2	2026-02-16	2026-03-27	30 days	
8	MS2 - Implementation planning	2026-02-16	2026-02-16	1 day	
4	Team Meeting 1	2026-02-16	2026-02-16	1 day	
5	Team Meeting 2	2026-03-02	2026-03-02	1 day	
9	MS2 - Implementation Design completed	2026-03-16	2026-03-20	5 days	
6	Team Meeting 3	2026-03-16	2026-03-16	1 day	
11	Milestone 3	2026-03-30	2026-04-22	18 days	
12	MS3 - Planning	2026-03-31	2026-03-31	1 day	
7	Team Meeting 4	2026-03-30	2026-03-30	1 day	
16	MS3 - Implementation Finished	2026-04-13	2026-04-16	4 days	
14	Team Meeting 5	2026-04-13	2026-04-13	1 day	
10	MS3 - Testing	2026-04-17	2026-04-20	2 days	
15	Project Finalization	2026-04-20	2026-04-22	3 days	
17	Team Meeting 6	2026-04-20	2026-04-20	1 day	

GitHub Link

https://github.com/aahall05/COP4533-Project-Group-6.git

Solutions

Problem Statement 1

1. $A = \begin{matrix} 12 & 1 & 5 & 3 & 16 \\ 4 & 4 & 13 & 4 & 9 \\ 6 & 8 & 6 & 1 & 2 \\ 14 & 3 & 4 & 8 & 10 \end{matrix}$
2. Profit for each sale is calculated using $(sell\ price) - (price\ paid)$
    a. Limitations:
        i. If bought and sold on the same day, profit is zero
        ii. The last stock can only be bought and sold on the last day
        iii. A stock bought on one day, can not be sold on any days prior
    b. For the first stock:
        i. Buying on day one, the profits for selling each subsequent day would be:
            1. $-11 \quad -7 \quad -9 \quad 4$
        ii. Buying on day two, the profits for selling each subsequent day would be:
            1. $4 \quad -2 \quad 15$
        iii. Buying on day three, the profits for selling each subsequent day would be:
            1. $-2 \quad 11$
        iv. Buying on day four, the profit for selling each subsequent day would be:
            1. 13
        v. Buying on day five, the profit would be 0 since it must be sold that day
    c. For the second stock:
        i. Buying on day one, the profits for selling each day would be:
            1. $0 \quad 9 \quad 0 \quad 5$
        ii. Buying on day two, the profits for selling each day would be:
            1. $9 \quad 0 \quad 5$
        iii. Buying on day three, the profits for selling each day would be:
            1. $-9 \quad -4$
        iv. Buying on day four, the profit for selling each day would be:
            1. 5
        v. Buying on day four, the profit would be 0 since it must be sold that day

d. For the third stock:
    i. Buying on day one, the profits for selling each subsequent day would be:
        1. $2 \quad 0 \quad -5 \quad -4$
    ii. Buying on day two, the profits for selling each subsequent day would be:
        1. $-2 \quad -7 \quad -6$
    iii. Buying on day three, the profits for selling each subsequent day would be:
        1. $-5 \quad -4$
    iv. Buying on day four, the profit for selling each subsequent day would be:
        1. -1
    v. Buying on day five, the profit would be 0 since it must be sold that day
e. For the fourth stock:
    i. Buying on day one, the profits for selling each day would be:
        1. $-11 \quad -10 \quad -6 \quad -4$
    ii. Buying on day two, the profits for selling each day would be:
        1. $1 \quad 5 \quad 7$
    iii. Buying on day three, the profits for selling each day would be:
        1. $4 \quad 6$
    iv. Buying on day four, the profit for selling each day would be:
        1. 2
    v. Buying on day four, the profit would be 0 since it must be sold that day

3. For each stock, the best day to buy and sell is: in the format (buy day, sell day)=profit
    a. Stock 1: $(2,5) = 15$
    b. Stock 2: $(1,3) = 9$
    c. Stock 3: $(1,2) = 2$
    d. Stock 4: $(2,5) = 7$
4. So the stock, and the days to buy and sell it, with the highest profit is stock one, bought on the second day and sold on the fifth day
    a. $(1,2,5,15)$

First we will find the best days to buy and sell each stock to maximize profit

Problem Statement 2

$$A = \begin{matrix} 25 & 30 & 15 & 40 & 50 \\ 10 & 20 & 30 & 25 & 5 \\ 30 & 45 & 35 & 10 & 15 \\ 5 & 50 & 35 & 25 & 45 \end{matrix}$$

I = 4 different stocks

J = 5 days

K = Constraint of maximum of 3 transactions

1. First we calculate potential profit for single-day interval.
    a. To determine the profit, simply subtract the earlier buy price from the later sell price. For Stock 1, we subtract the Day 1 price $25 at A{1,1} from the Day 2 price $30 at A{1,2}, resulting in a profit of $5.

- **Day 1 → 2**
    - Stock 1: $30 – $25 = $5
    - Stock 2: $20 – $10 = $10
    - Stock 3: $45 - $30 = $15
    - Stock 4: $50 - $5 = $45
    - **Optimal:** Stock 4 (Profit: 45)
- **Day 2 → Day 3**
    - Stock 1: $15 - $30 = -$15
    - Stock 2: $30 - $20 = $10
    - Stock 3: $35 - $45 = -$10
    - Stock 4: $35 - $50 = -$15
    - **Optimal:** Stock 2 (Profit: 10)
- **Day 3 → Day 4**
    - Stock 1: $40 - $15 = $25
    - Stock 2: $25 - $30 = -$5
    - Stock 3: $10 - $35 = -$25
    - Stock 4: $25 - $35 = -$10
    - **Optimal**: Stock 1 (Profit: 25)
- **Day 4 → Day 5**
    - Stock 1: $50 - $40 = 10
    - Stock 2: $5 - $25 = -$20$
    - Stock 3: $15 - $10 = 5
    - Stock 4: $45 - $25 = $20
    - **Optimal:** Stock 4 (Profit: 20)

    b. Now that we've done the calculations for each day, we can select the optimal transactions which are the most profitable:

i. **Day 1-2:** Profit **45** (Stock 4)
ii. **Day 2-3:** Profit **10** (Stock 2)
iii. **Day 3-4:** Profit **25** (Stock 1)
iv. **Day 4-5:** Profit **20** (Stock 4)

c. We are limited to 3 transactions per the problem statement (k = 3), so we just chose the three that are the most profitable, which gives a **total profit of $90.**

2. Applying these methods to Matrix A, we get this transaction sequence:
   a. Stock 4:
      i. Buy Day 1, Sell Day 2 = $45 Profit
   b. Stock 1:
      i. Buy Day 3, Sell Day 4 = $25 Profit
   c. Stock 4: Buy Day 4, Sell Day 5 = $20 Profit

3. This can be expressed in the (I, j, l) format given as:
   a. [(4, 1, 2), (1, 3, 4), (4, 4, 5)]

Problem Statement 3

1. $A = \begin{matrix} 7 & 1 & 5 & 3 & 6 & 8 & 9 \\ 2 & 4 & 3 & 7 & 9 & 1 & 8 \\ 5 & 8 & 9 & 1 & 2 & 3 & 10 \\ 9 & 3 & 4 & 8 & 7 & 4 & 1 \\ 3 & 1 & 5 & 8 & 9 & 6 & 4 \end{matrix}$ c =2
2. With c = 2, in order to perform two transactions, the first stock must be bought on day one or two
   a. If a stock is bought on day one, it must be sold on day two or day three
   b. If a stock is bought on day two, it must be sold on day three.
3. This gives us three options for the first stock:
   a. (n, 1, 2)
   b. (n, 1, 3)
   c. (n, 2, 3)
4. The profits for option a:
   a. (1, 1, 2) = 1 – 7 = -6
   b. (2, 1, 2) = 4 – 2 = 2
   c. (3, 1, 2) = 8 - 5 = 3
   d. (4, 1, 2) = 3 – 9 = -6
   e. (5, 1, 2) = 1 – 3 = -2

5. The profits for option b:
    a. (1, 1, 3) = 5 – 7 = -2
    b. (2, 1, 3) = 3 – 2 = 1
    c. (3, 1, 3) = 9 - 5 = 4
    d. (4, 1, 3) = 4 – 9 = -5
    e. (5, 1, 3) = 5 – 3 = 2
6. The profits for option c
    a. (1, 2, 3) = 5 - 1 = 4
    b. (2, 2, 3) = 3 - 4= -1
    c. (3, 2, 3) = 9 - 8 = 1
    d. (4, 2, 3) = 4 - 3 = 1
    e. (5, 2, 3) = 5 - 1 = 4
7. The max profit of all these options is 4. This is achieved through multiple trades: (3, 1, 3), (1, 2, 3), (5, 2, 3). We will choose the first available option: (3, 1, 3)
8. With that trade being our first, we cannot buy another stock for c = 2 days.
9. This means our second trade must be bought on day 6 and sold on day 7
10. The possible profits for those specific buy/sell days are:
    a. (1, 6, 7) = 9 - 8 = 1
    b. (2, 6, 7) = 8 - 1 = 7
    c. (3, 6, 7) = 10 - 3 = 7
    d. (4, 6, 7) = 1 - 4 = -3
    e. (5, 6, 7) = 6 - 4 = 2
11. The maximum profit possible for that second day is 7 and can be achieved using multiple trades: (2, 6, 7), (3, 6, 7)
12. Taking the first trade, we have (2, 6, 7)
13. The two trades that maximize profit are
    a. (3, 1, 3) = 4
    b. (2, 6, 7) = 7
14. So adding these together we get 11 profit
15. The best trade sequence is [(3, 1, 3), (2, 6, 7)] = 11